

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5:
Measure network performance using key metrics with simulation tools
(BL4,BL5)

Assessment Tool Weightage: 40%

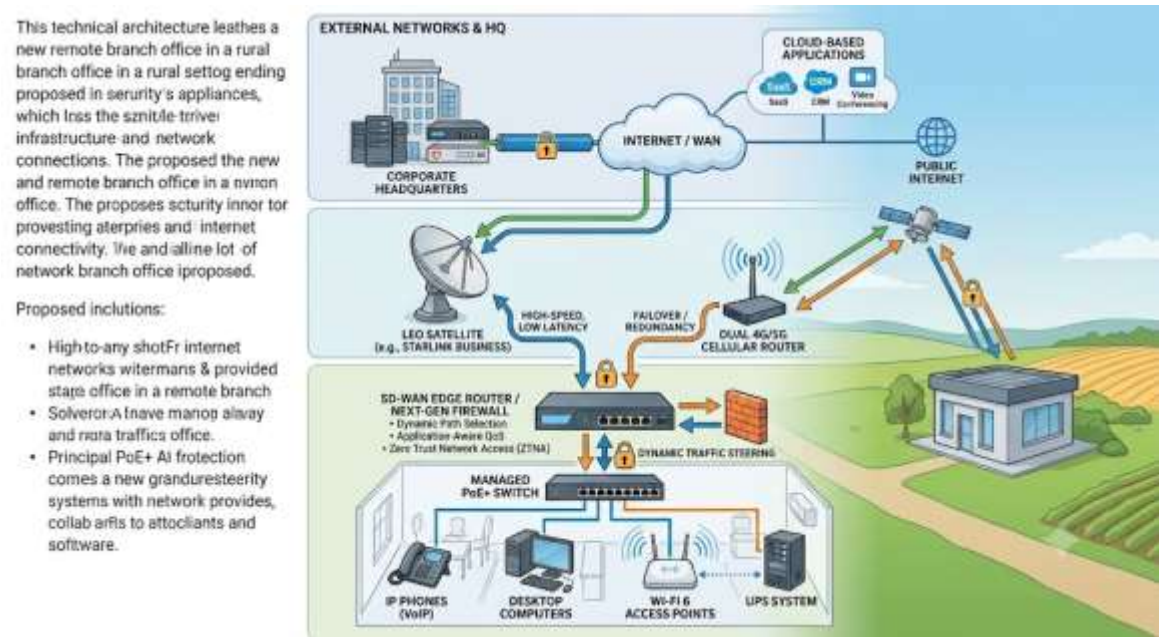
QUESTIONS: 3 Total Marks:30

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability

To connect a remote branch office while meeting requirements for cloud apps, video conferencing, secure HQ communication, and tight budget constraints, a **hybrid wireless WAN architecture** paired with an **SD-WAN deployment** is the optimal design.



Recommended Network Architecture

1. Internet Connectivity: Hybrid LEO Satellite + Dual 4G/5G Cellular

- **Primary Connection:** Low Earth Orbit (LEO) Satellite (e.g., Starlink Business). Provides high-speed throughput (100–220+ Mbps) with low latency (25–40 ms), sufficient for video conferencing and cloud SaaS applications.

- **Secondary/Failover Connection:** Dual-SIM 4G LTE / 5G Enterprise Cellular Router. Serves as a backup link to guarantee zero downtime during weather disruptions or satellite satellite outages.

2. Core Networking Devices

- **SD-WAN Edge Router / Firewall:** A unified appliance (e.g., Fortinet FortiGate 60F or Cisco Meraki MX Series). Manages dynamic WAN bonding, real-time traffic steering, and dynamic failover.
- **Managed PoE+ Switch:** A 24-port Managed Layer 2/3 Switch (e.g., Cisco Catalyst 1000 or Aruba CX 6000) to power IP phones, access points, and desktops while providing VLAN segmentation.
- **Enterprise Wi-Fi Access Points:** Dual-band Wi-Fi 6 APs (e.g., Ubiquiti Unifi APs or Meraki MR) deployed for seamless coverage across the office.
- **Uninterruptible Power Supply (UPS):** A 1500VA Line-Interactive UPS to protect networking equipment against erratic rural power grids.

3. Security Mechanisms

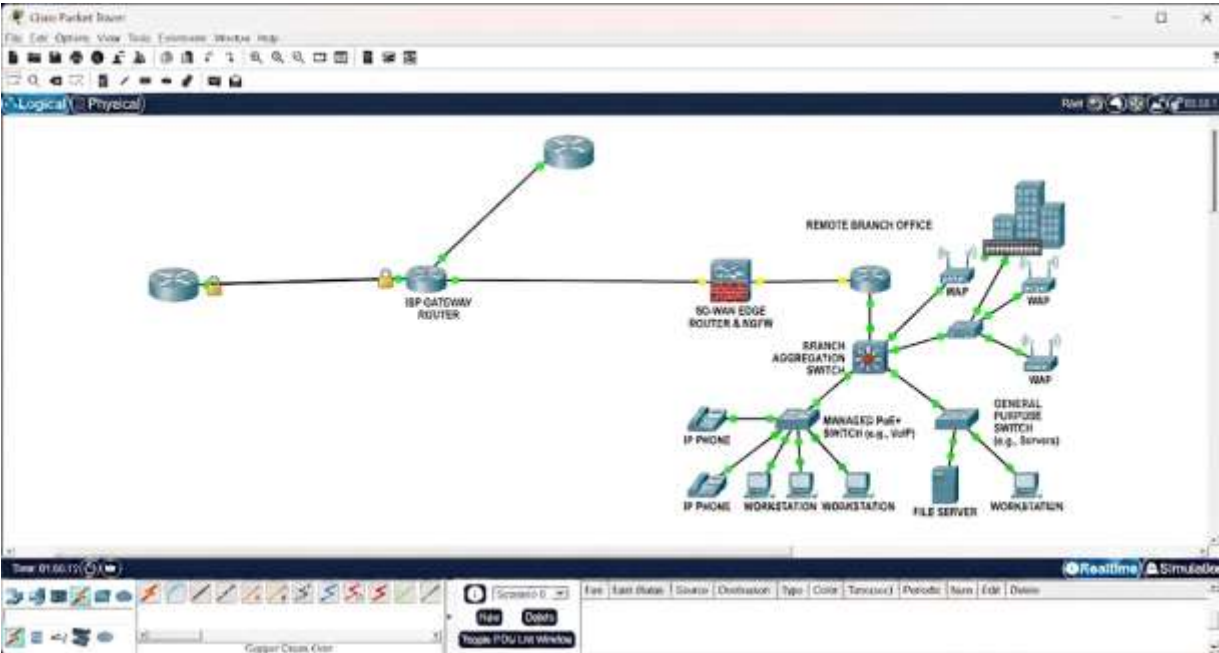
- **IPsec VPN Tunneling:** An encrypted Site-to-Site IPsec VPN tunnel with AES-256 encryption to connect the branch directly to corporate HQ or primary cloud data centers.
- **SD-WAN Application-Aware QoS:** Traffic shaping rules prioritize VoIP and real-time video streams over non-critical background traffic, mitigating potential jitter on satellite links.
- **Next-Generation Firewall (NGFW) & SASE:** Unified Threat Management (UTM) offering Intrusion Prevention (IPS), Antivirus, Web Filtering, and Zero Trust Network Access (ZTNA) directly on the edge router.

Technical Justification Matrix

Parameter	Recommended Solution	Justification
Bandwidth	LEO Satellite + Cellular Aggregation	LEO satellite solves the high-bandwidth demand (100+ Mbps) for cloud apps, while 5G/LTE handles burst traffic or dedicated VoIP channels.
Reliability	Dual-WAN Failover via SD-WAN	Seamless sub-second failover guarantees uninterrupted operations . If the satellite connection drops due to storm conditions, traffic immediately fails over to cellular.
Cost Efficiency	LEO + Cellular over Dedicated Fiber/GEO	Traditional leased lines or GEO satellites carry exorbitant setup costs and recurring fees. LEO + Cellular delivers enterprise performance at a fraction of the CAPEX/OPEX.

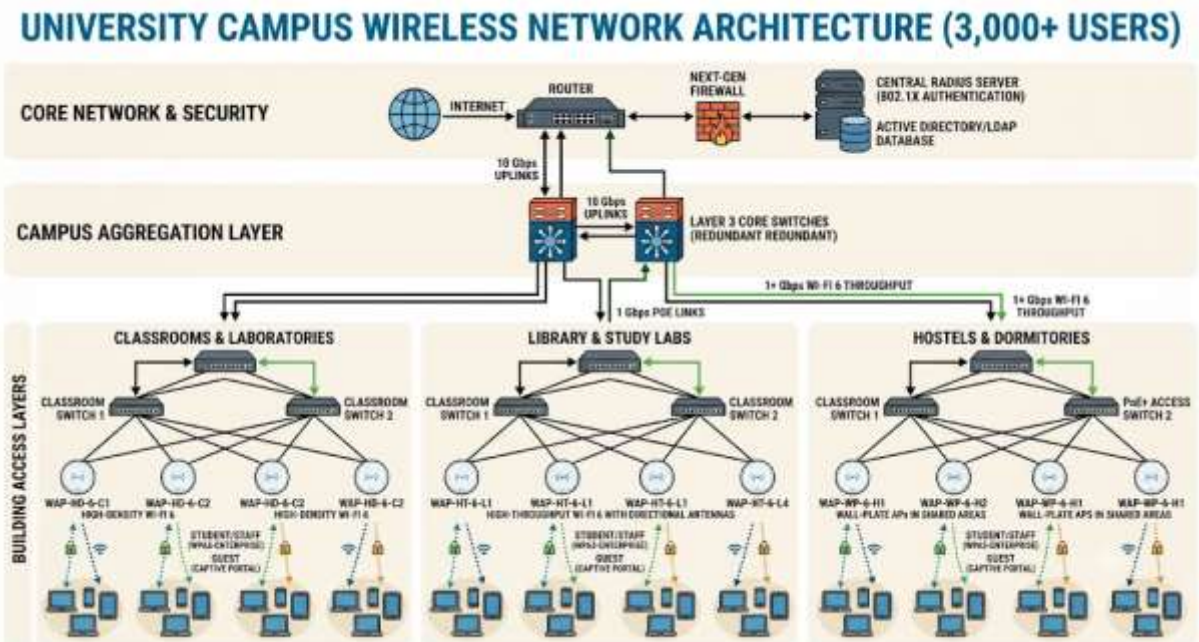
Scalability	Cloud-Managed SD-WAN Infrastructure	Centralized cloud controllers allow network engineers to push firewall policies, routing updates, and security patches remotely without on-site technical staff.
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Output



- A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.

To deliver high-performance, secure, campus-wide Wi-Fi for 3,000+ students under strict budget constraints, the design relies on **high-density spatial planning, dual-band channel optimization, centralized cloud/virtual management, and 802.1X enterprise security.**



Core Recommendations & Network Architecture

1. Frequency Band Selection & Channel Management

- **Dual-Band (2.4 GHz & 5 GHz / 6 GHz ready):**
 - **2.4 GHz (20 MHz channels):** Used strictly for long-range coverage in low-density open outdoor spaces (using non-overlapping Channels 1, 6, 11). Bandwidth-heavy apps are throttled here.
 - **5 GHz (20/40 MHz channels):** Deployed across high-density indoor areas (Classrooms, Labs, Library). Dynamic Frequency Selection (DFS) channels are enabled to expand available non-overlapping spectrum.
- **Band Steering:** Configured on all Access Points (APs) to automatically push dual-band client devices onto the faster 5 GHz band, keeping 2.4 GHz clear for legacy devices.

2. Strategic Access Point Placement & Density Control

- **Classrooms & Lecture Halls (High Capacity):** High-density Wi-Fi 6 (802.11ax) 4x4 MU-MIMO APs mounted centrally on ceilings. AP power levels are turned down to decrease cell size, preventing co-channel interference (CCI) while enabling spatial reuse.

Network Design Parameters & Technical Trade-offs

- **Libraries & Study Labs (High Throughput):** Standard 2x2 Wi-Fi 6 APs placed near high-occupancy seating grids with directional antennas to focus signals on target areas.

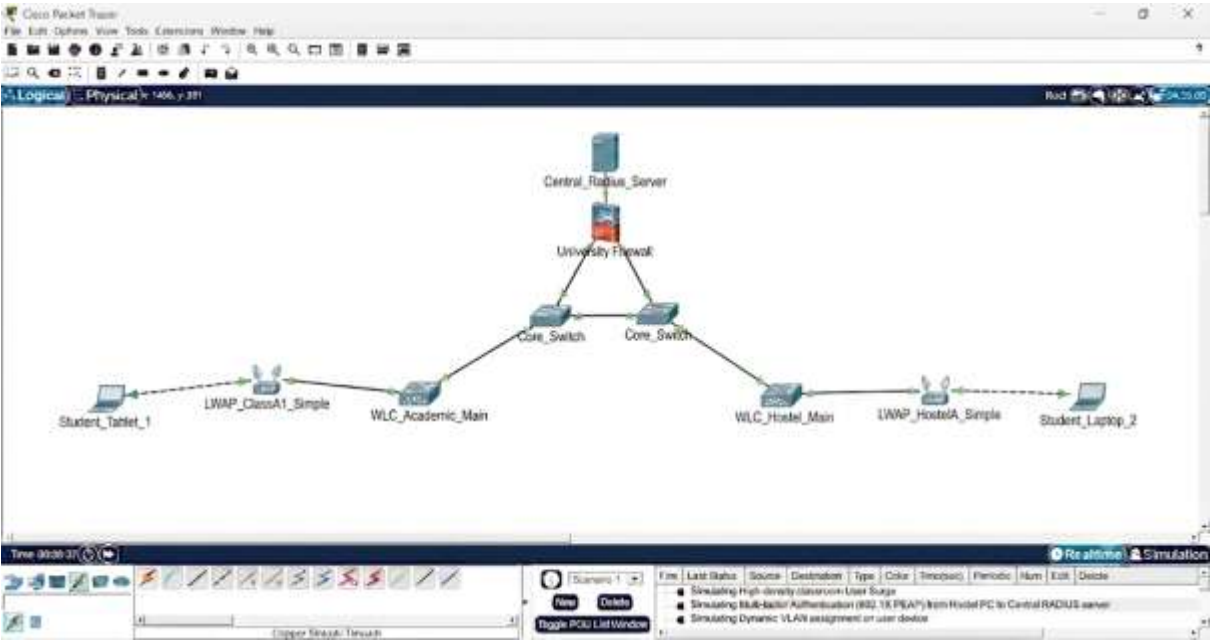
Parameter	Proposed Design Strategy	Cost & Performance Balance
Coverage vs. Cost	Wi-Fi 6 (802.11ax) + 802.11ax Target Wake Time (TWT)	Wi-Fi 6 handles 4x more concurrent devices per AP than Wi-Fi 5. Buying fewer, higher-capacity APs reduces total CAPEX while meeting the 3,000-user load.
Interference Control	Dynamic Channel and Power Adaptation (RRM/ARM)	Automated RF management dynamically adjusts AP transmission power and channels to eliminate co-channel interference without manual tuning.
Traffic Management	VLAN Segmentation & Airtime Fairness	Network divided into VLANs (Students, Faculty, IoT, Guests). Airtime Fairness prevents slow/legacy devices from hogging wireless bandwidth.
Management	Cloud-Managed or Virtual Controller Architecture	Eliminates expensive hardware controllers by using cloud management (e.g., Cisco Catalyst Center, Aruba Central, or Ubiquiti UniFi Hosted) with a single-pane-of-glass dashboard.

- **Hostels & Dormitories (High Wall-Loss Areas):** Wall-plate APs placed in hallways or shared study lounges rather than every individual room, leveraging hallway coverage to serve 4–6 rooms per AP within budget limits.
- **Outdoor Pathways:** Weatherproof IP67-rated APs with directional sector antennas positioned on central building exteriors to bridge outdoor coverage without deploying dedicated poles.

3. Enterprise Security & Authentication

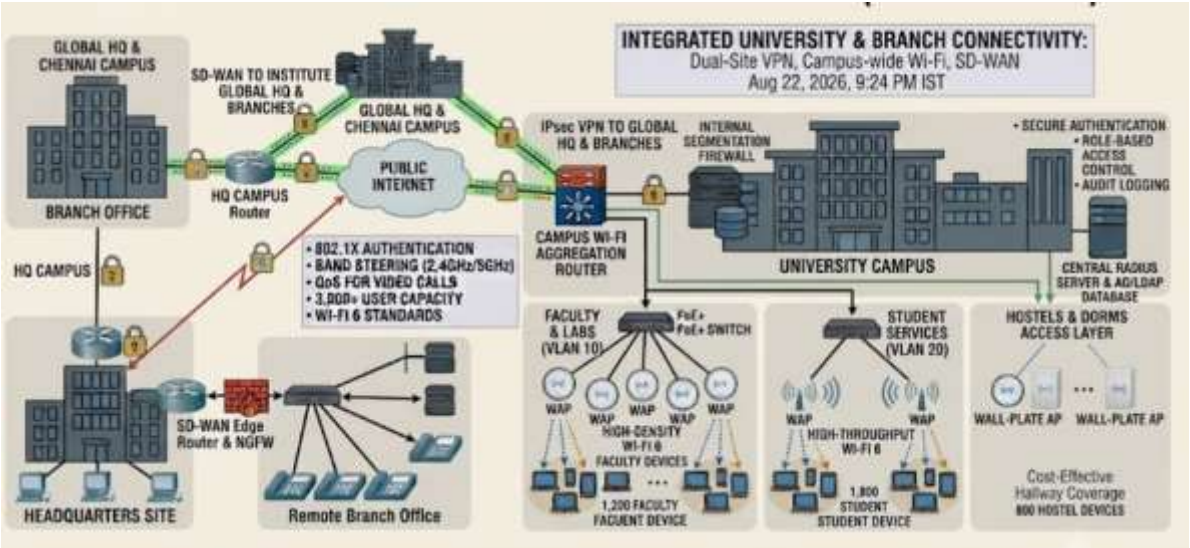
- Primary Student/Staff Network (WPA3-Enterprise / 802.11r):**
 - Authentication:** RADIUS Server paired with the university’s central Active Directory / LDAP database via **WPA3-Enterprise (AES-CCMP/GCMP encryption)**.
 - Seamless Roaming:** IEEE 802.11r (**Fast BSS Transition**) and 802.11k/v enable students to walk across campus during video calls without dropping connections.
- Guest Network:** Captive portal with time-limited OTP/email authentication, strictly isolated into an internet-only VLAN with client isolation enabled.

Output



3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

To meet strict financial compliance standards (e.g., PCI-DSS, SOC 2) while protecting sensitive customer data and maintaining high network throughput on a controlled budget, the network architecture utilizes a **Zero-Trust, Multi-Tiered Segmented Design**.



Key Architectural Components

1. Network Segmentation (VLANs & Micro-segmentation)

- **Demilitarized Zone (DMZ):** Hosts public-facing web servers and online banking portals. Isolated from the internal network.
- **Corporate / Employee Zone:** Restricts general staff workstations. Access to databases is blocked by default.
- **Database & Core Processing Zone (PCI-DSS Zone):** Holds customer PII and transaction records. Isolated via Layer 3 VLANs and Next-Generation Firewall (NGFW) micro-segmentation.
- **Management Zone:** Strictly reserved for network administrators using dedicated Jump Hosts and Out-of-Band (OOB) management.

2. Firewall Deployment Strategy

- **Perimeter NGFW:** Positioned at the Internet edge to filter inbound/outbound traffic, block IP reputation risks, and handle DDoS mitigation.
- **Internal Segmentation Firewall (ISFW):** Positioned between the DMZ, Corporate, and Sensitive Database zones. Inspects inter-zone traffic (East-West traffic) to prevent lateral movement if a corporate device is compromised.

3. Authentication & Access Control

- **Identity-Driven Access:** WPA3-Enterprise and 802.1X Port Security via RADIUS/TACACS+.
- **Multi-Factor Authentication (MFA):** Enforced for all internal database access, admin logins, and remote VPN connections.
- **Role-Based Access Control (RBAC):** Principle of Least Privilege (PoLP) ensures employees access only data required for their specific role.

4. Intrusion Detection & Prevention (IDS/IPS)

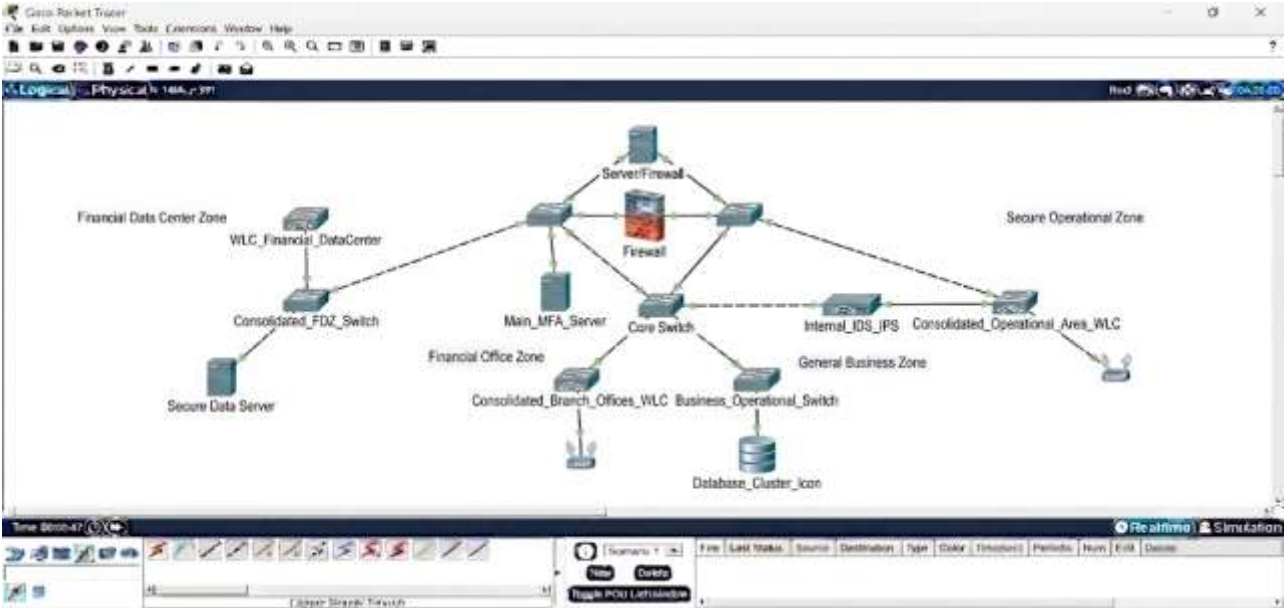
- **Inline Next-Gen IPS:** Integrated within the NGFW/ISFW to perform deep packet inspection (DPI) on traffic heading toward sensitive databases without adding latency.
- **Host-Based Detection (EDR):** Deployed on endpoints to monitor process execution and block ransomware at the device level.
- **Centralized Logging (SIEM):** Aggregates logs across firewalls, switches, and servers for automated threat detection and compliance auditing.

Trade-off & Constraint Justification

Constraint	Architectural Solution	Justification
Security	Zero-Trust & ISFW Micro-segmentation	Prevents lateral movement; even if an employee workstation is infected, the hacker cannot reach the backend banking database.

Performance	ASIC-Accelerated NGFWs & Hardware IPS	Utilizing hardware-accelerated appliances handles deep packet inspection at line-rate speed (10Gbps+), preventing network bottlenecks.
Compliance	Strict Isolation, MFA, & Centralized SIEM Logging	Fulfills core requirements for PCI-DSS, GLBA, and SOC 2 regarding access controls and immutable audit trails.
Budget	Integrated NGFW/IPS & Virtualized VLANs	Consolidates IDS/IPS features directly into the firewall to avoid purchasing standalone IPS hardware, maximizing ROI.

Output



RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand